DevOps Capstone Project:

Deployment of Online Food Ordering and Delivery Application

Submitted to - Vikul Sir

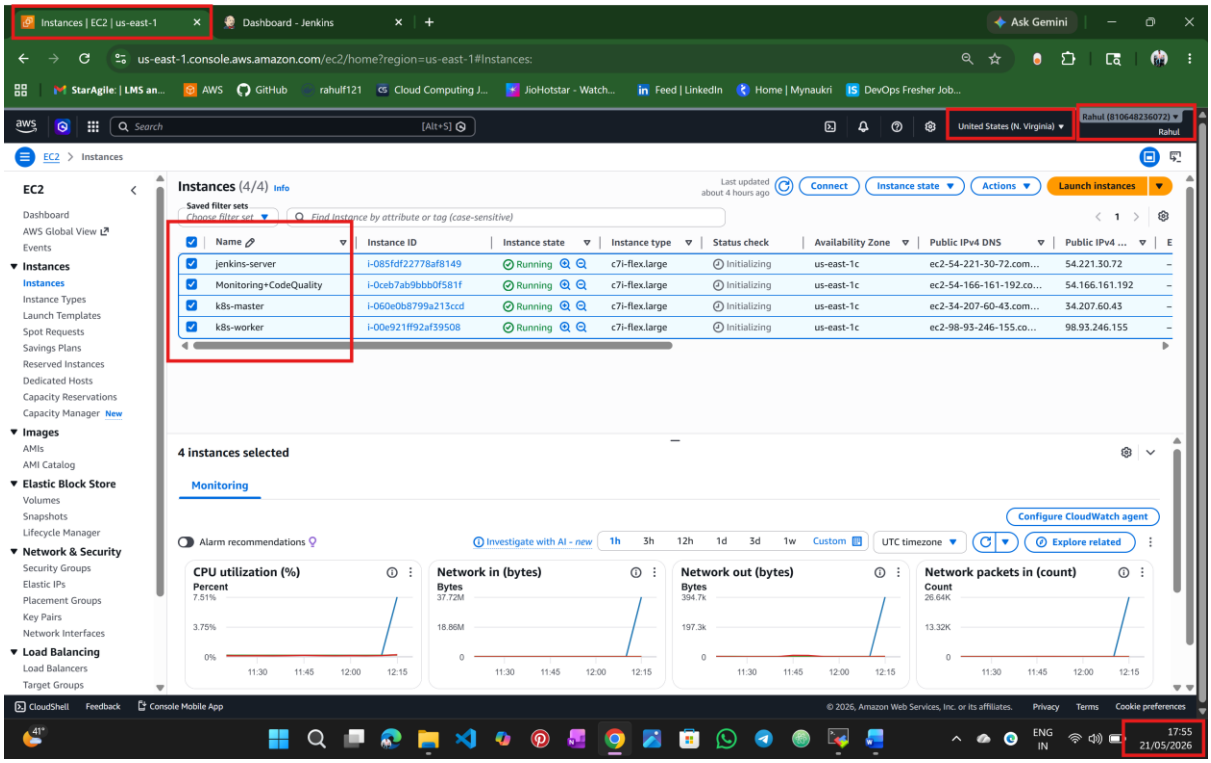
Date of Submission – 27/MAY/2026

Submitted by – Rahul Nalathati

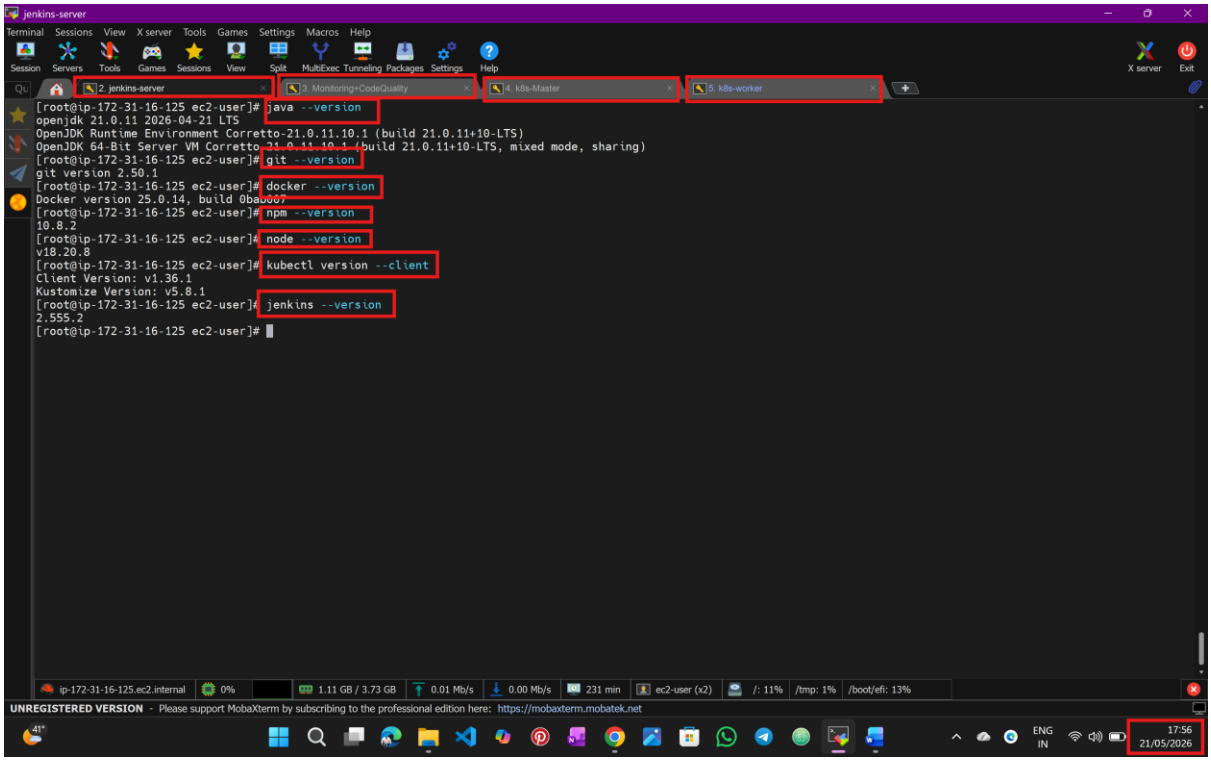
Project Overview:

Zomato is a leading online food delivery and restaurant discovery platform. To handle increasing user demand and improve deployment efficiency, the company is transitioning from a monolithic architecture to a microservices-based architecture using DevOps practices. The goal is to implement a CI/CD pipeline, containerized deployments, and real-time monitoring to ensure high availability, faster releases, and automated scalability. AWS will be used as the primary cloud provider for hosting and deployment.

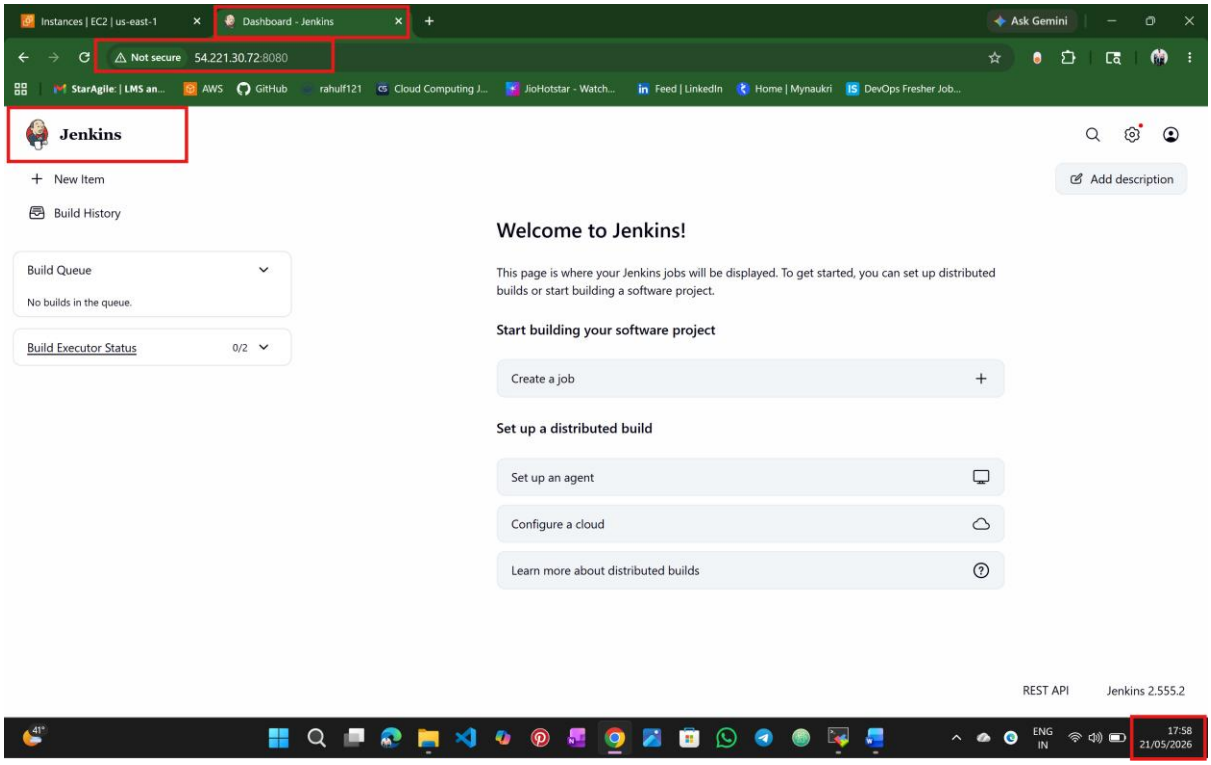
Step:1 Login into AWS console and Create 4 ec2 instances



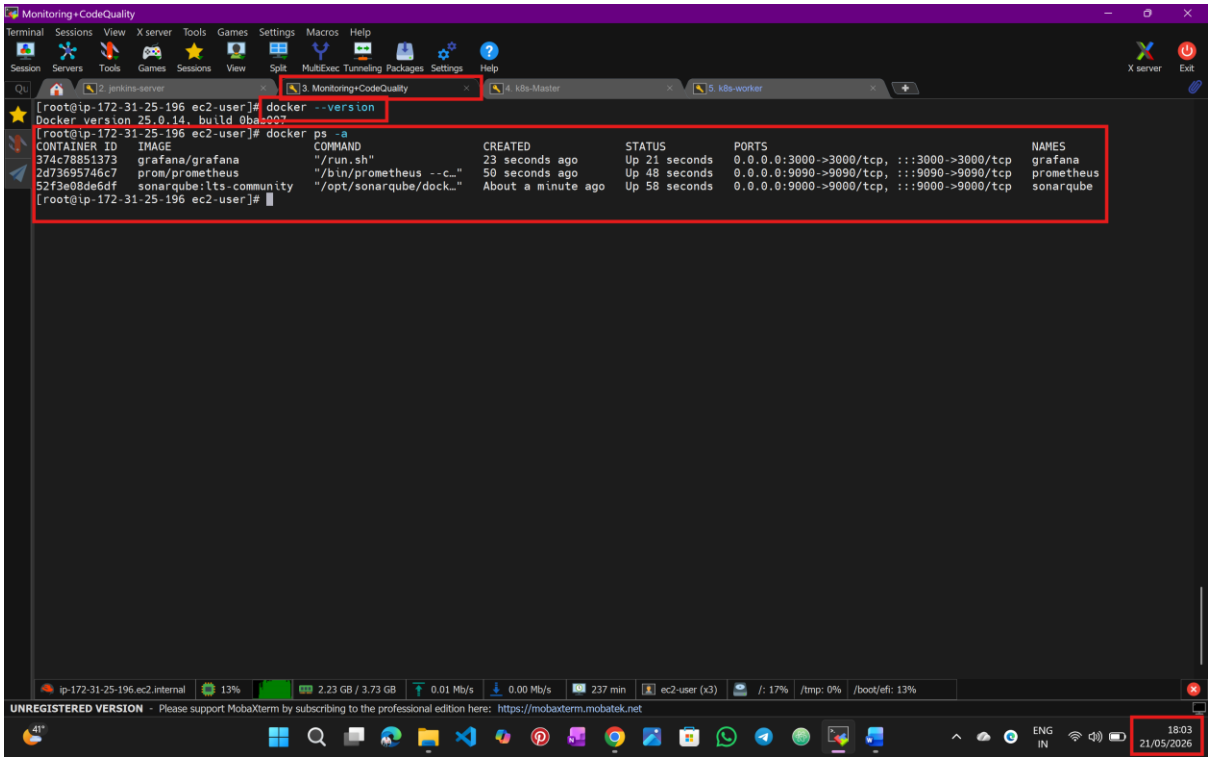
Step:2 connect all with SSH via MobaXterm and install prerequisite in all of them.



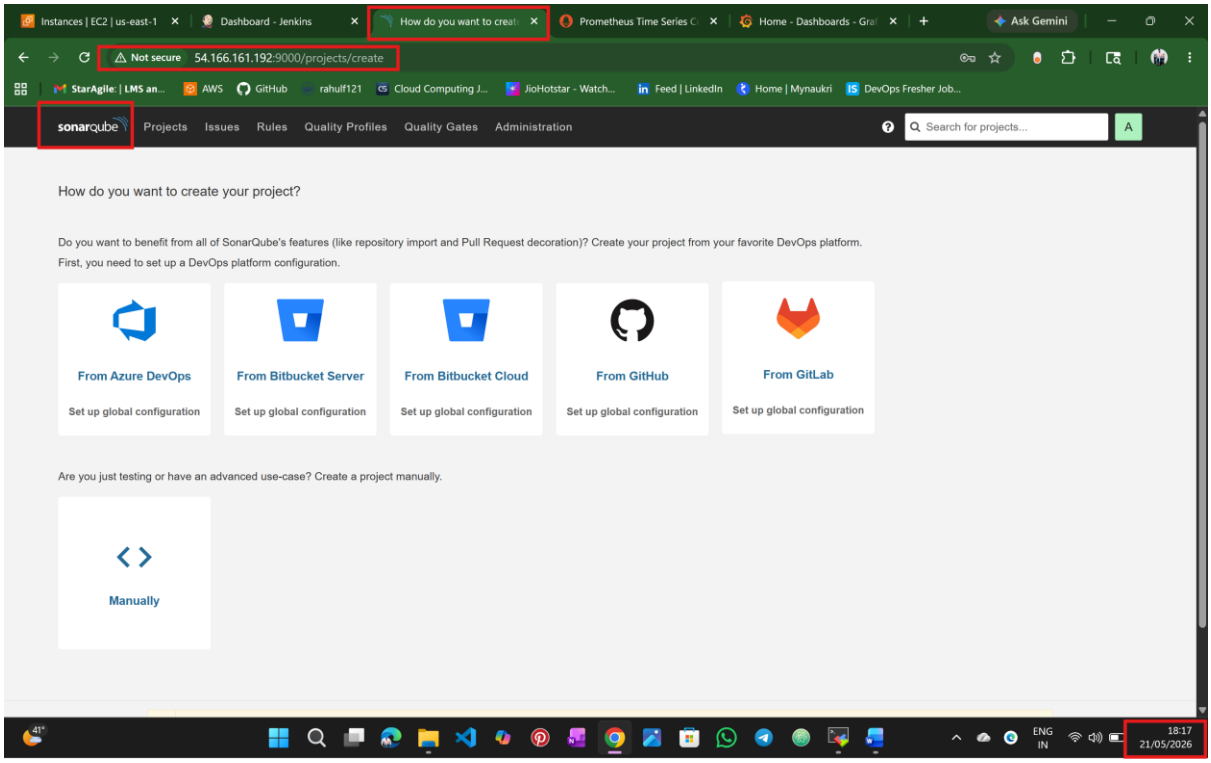
Step:3 Access the Jenkins website

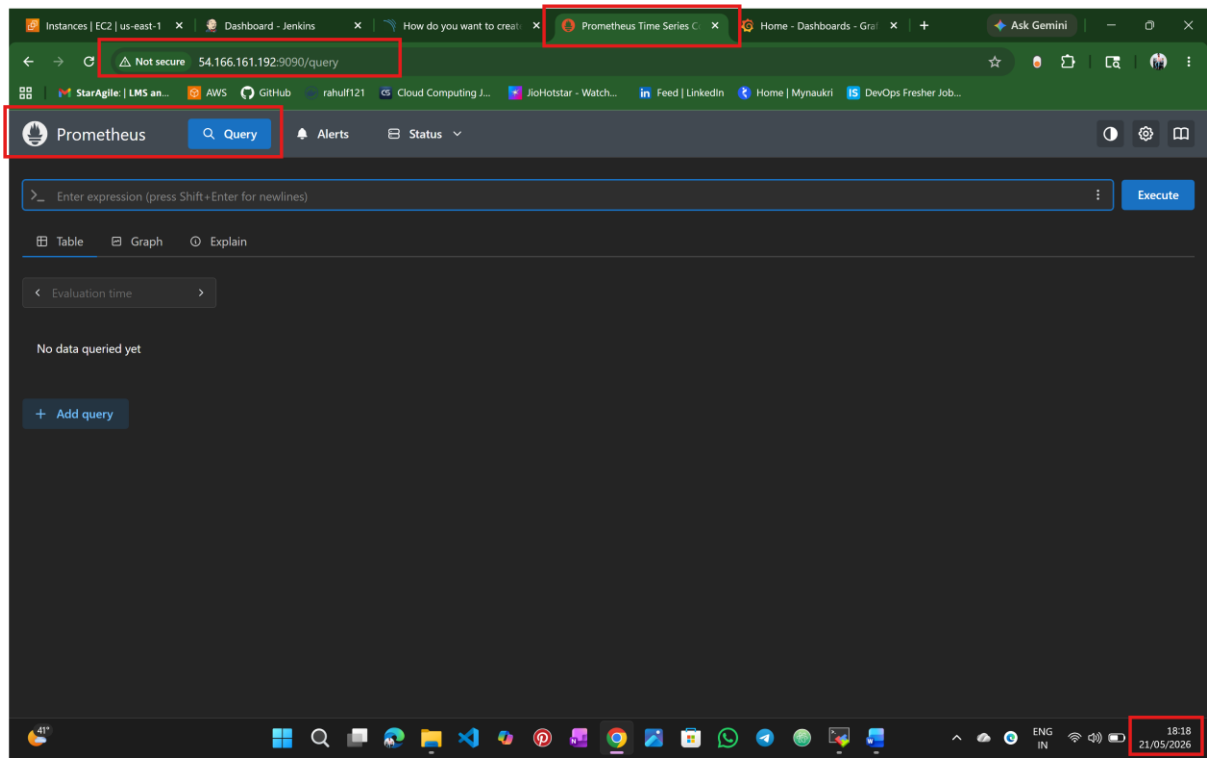


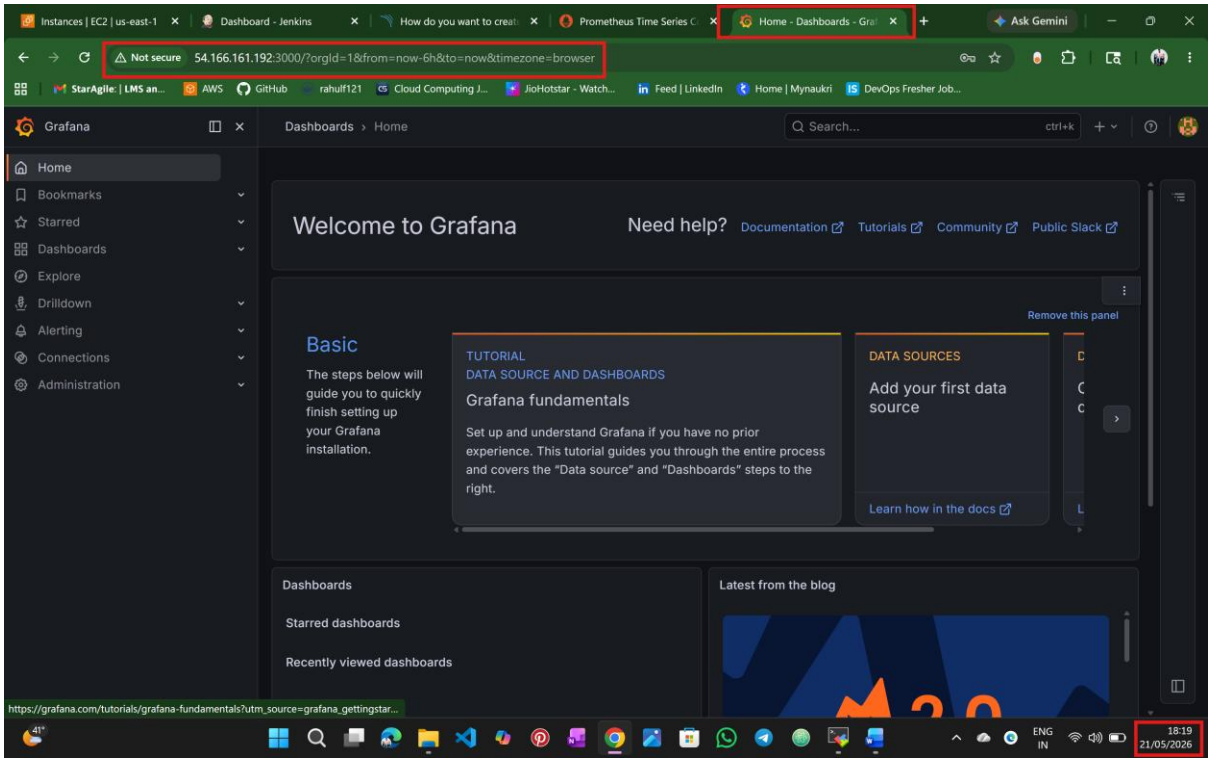
Step:4 install SonarQube, Prometheus and Grafana via Docker run



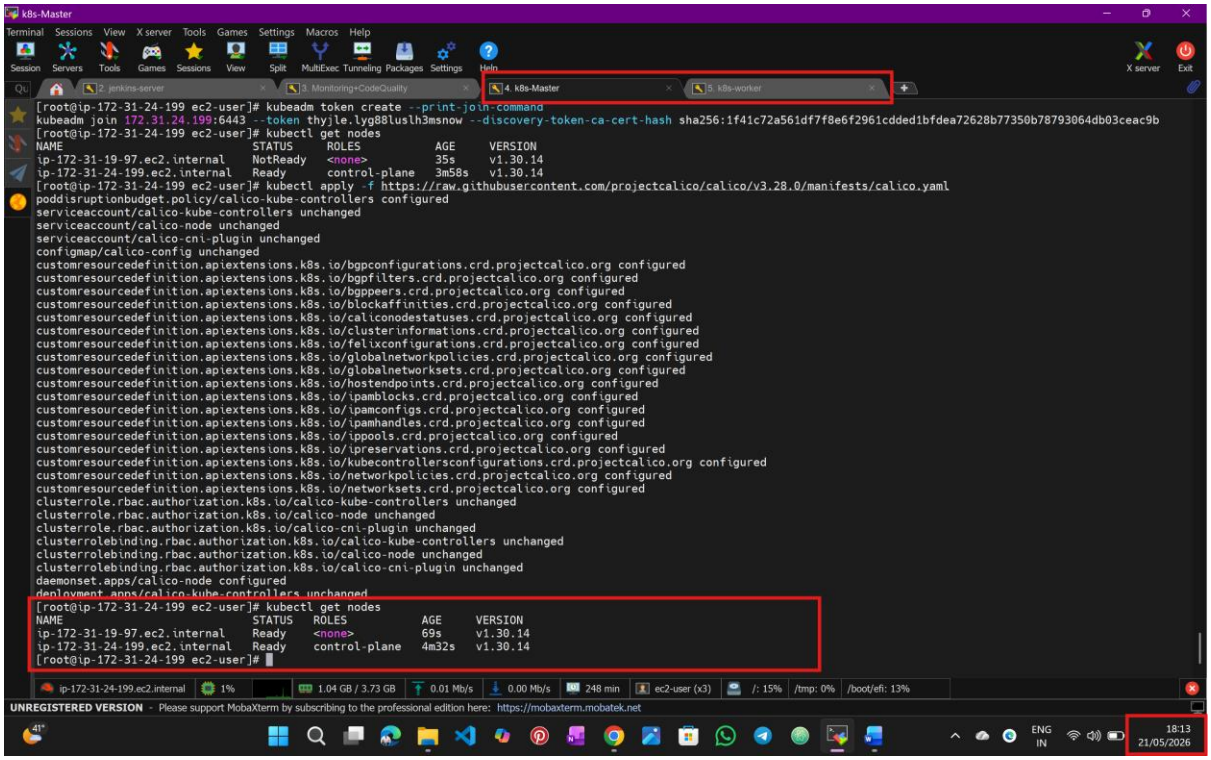
Step: 5 Access the SonarQube website, Prometheus and Grafana



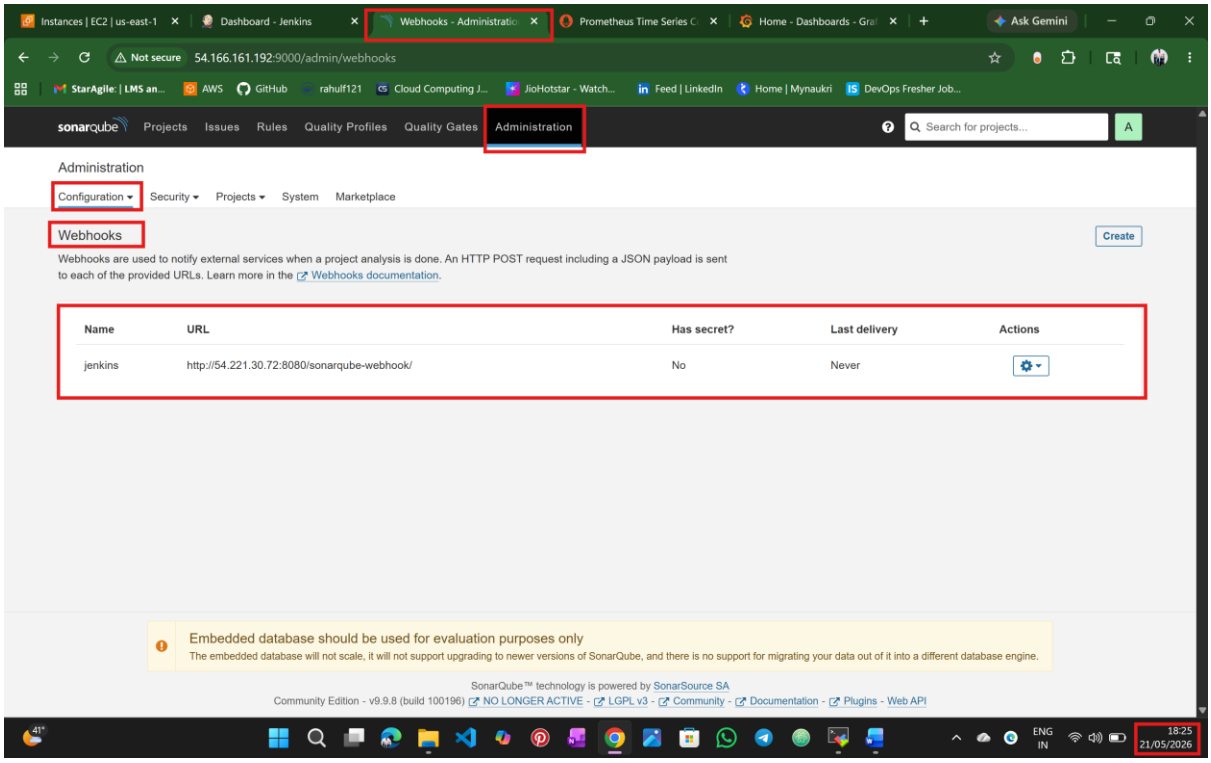

Not secure 54.166.161.192:9090/query
Prometheus
Query
Alerts
Status
Enter expression (press Shift+Enter for newlines)
Execute
Table
Graph
Explain
Evaluation time
No data queried yet
Add query


Not secure 54.166.161.192:3000/?orgId=1&from=now-6h&to=now&timezone=browser
Grafana
Dashboards › Home
Search...
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Connections
Administration
Welcome to Grafana
Need help?
Documentation
Tutorials
Community
Public Slack
Remove this panel
Basic
The steps below will guide you to quickly finish setting up your Grafana installation.
TUTORIAL
DATA SOURCE AND DASHBOARDS
Grafana fundamentals
Set up and understand Grafana if you have no prior experience. This tutorial guides you through the entire process and covers the "Data source" and "Dashboards" steps to the right.
DATA SOURCES
Add your first data source
Learn how in the docs
Dashboards
Starred dashboards
Recently viewed dashboards
Latest from the blog
https://grafana.com/tutorials/grafana-fundamentals?utm_source=grafana_gettingstar...

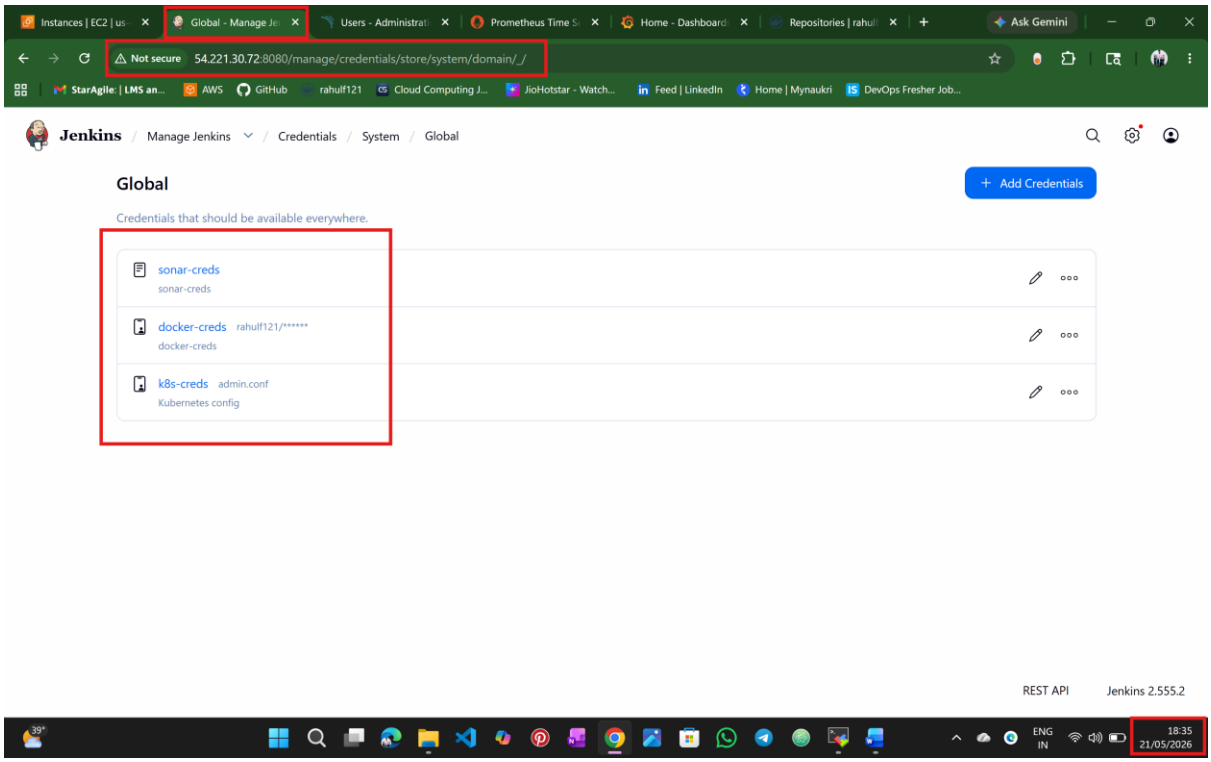
Step:6 connect K8s master and k8s worker nodes



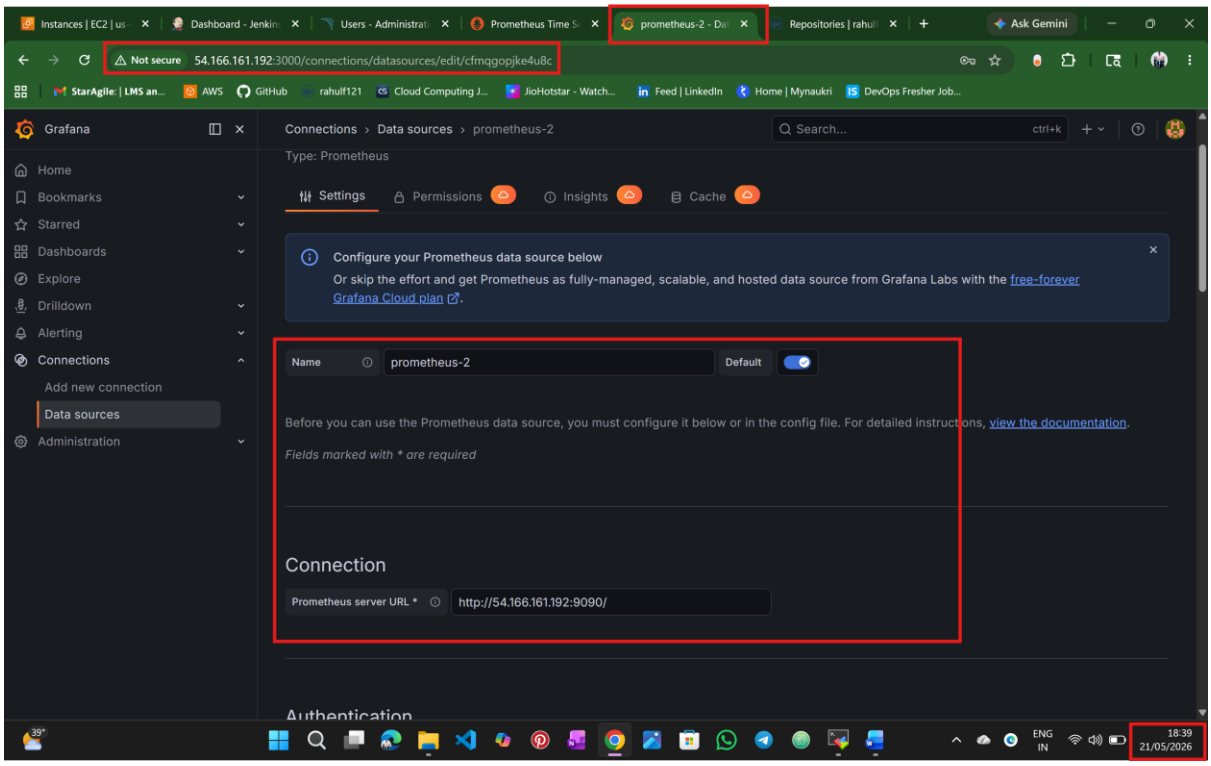
Step:7 connect SonarQube and Jenkins using webhook



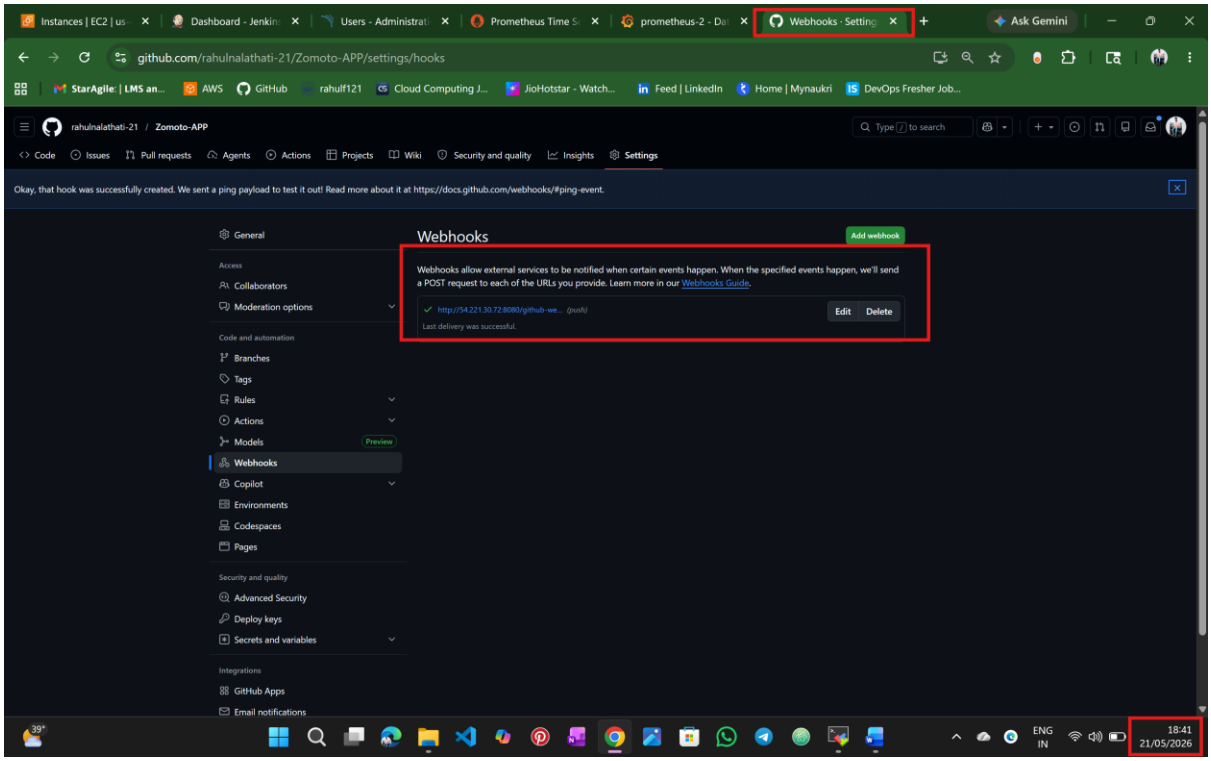
Step:8 store the credentials in jenkins



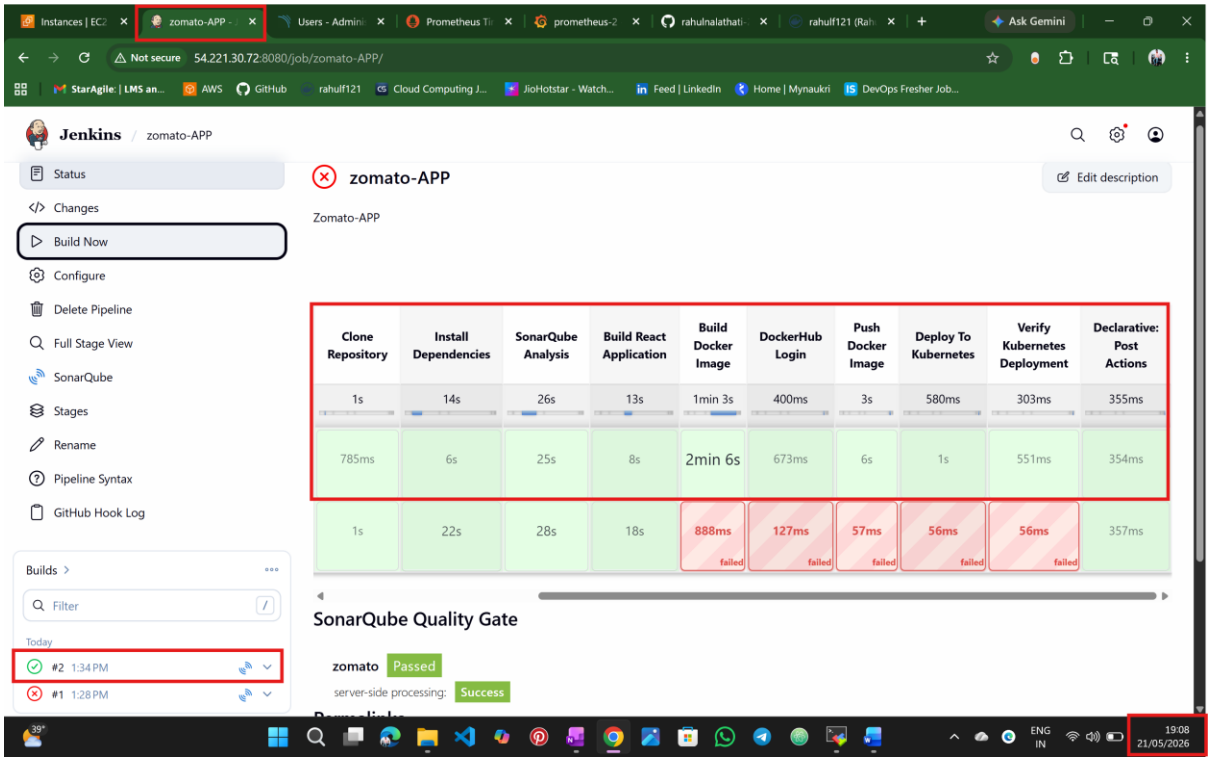
Step:9 connect Prometheus and Grafana



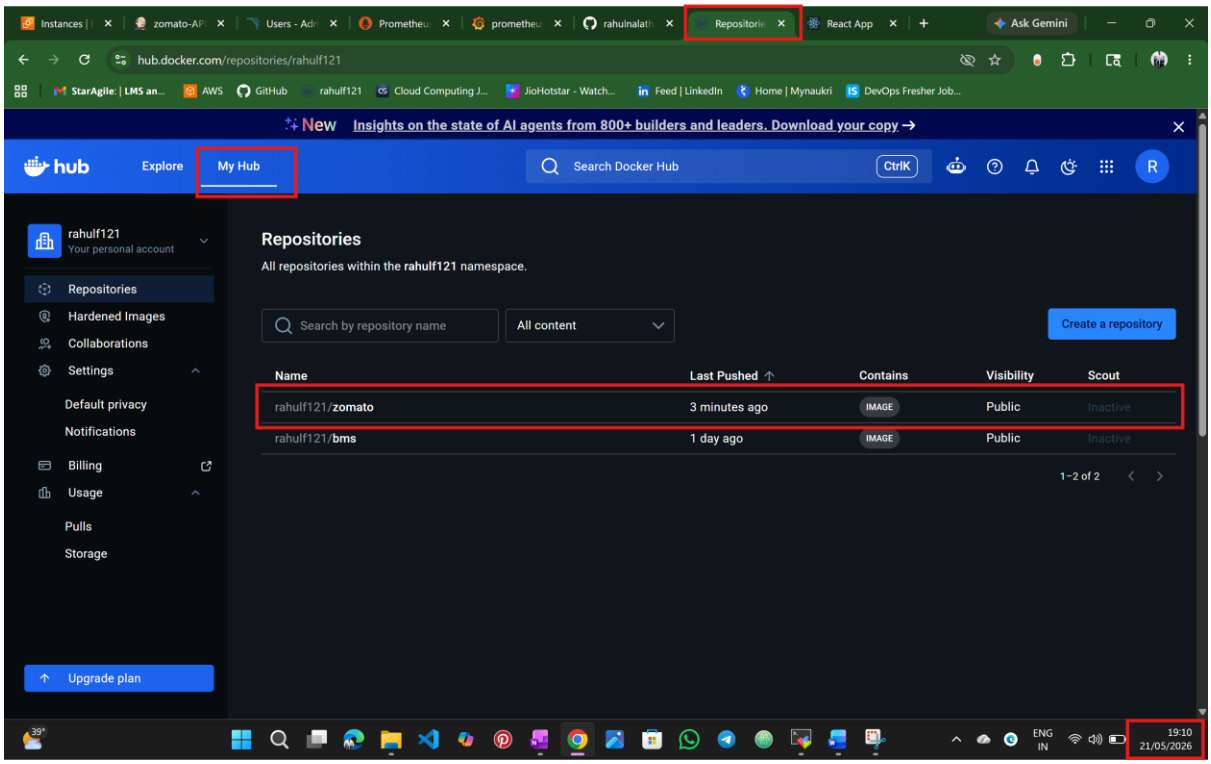
Step:10 connect git hub and Jenkins using webhook



Step:11 build the pipeline

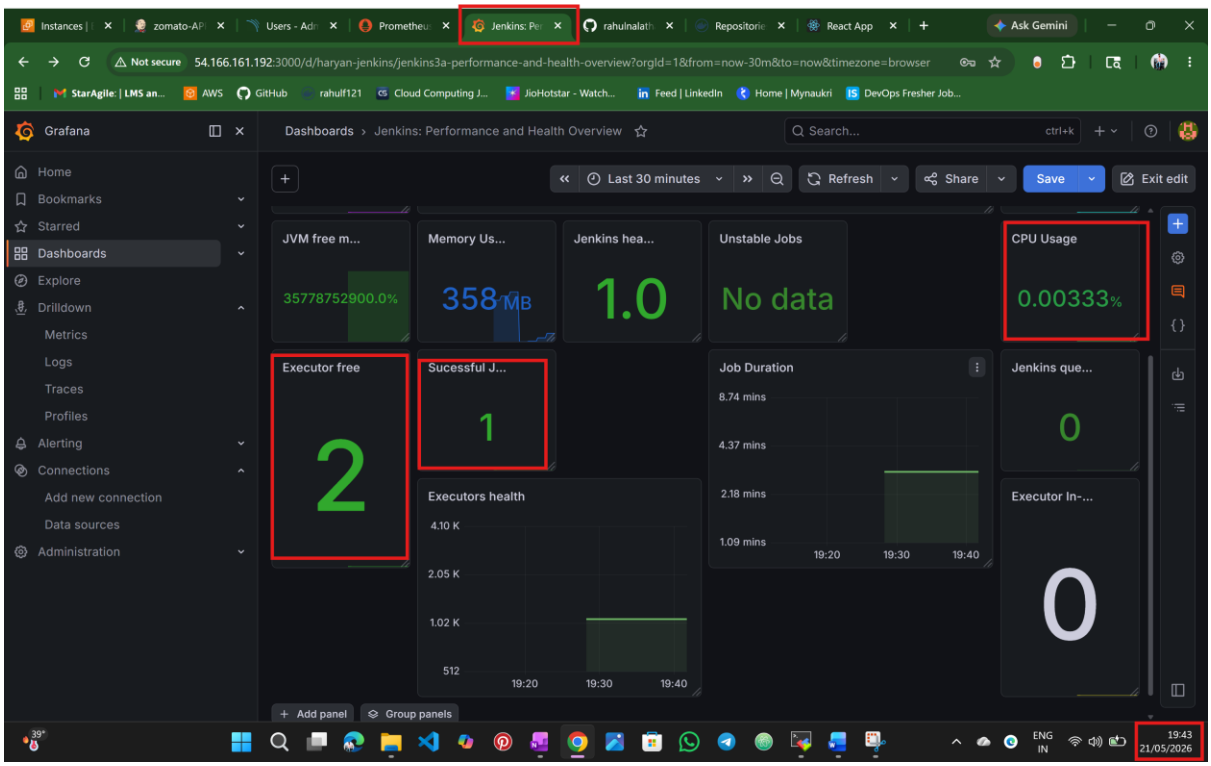


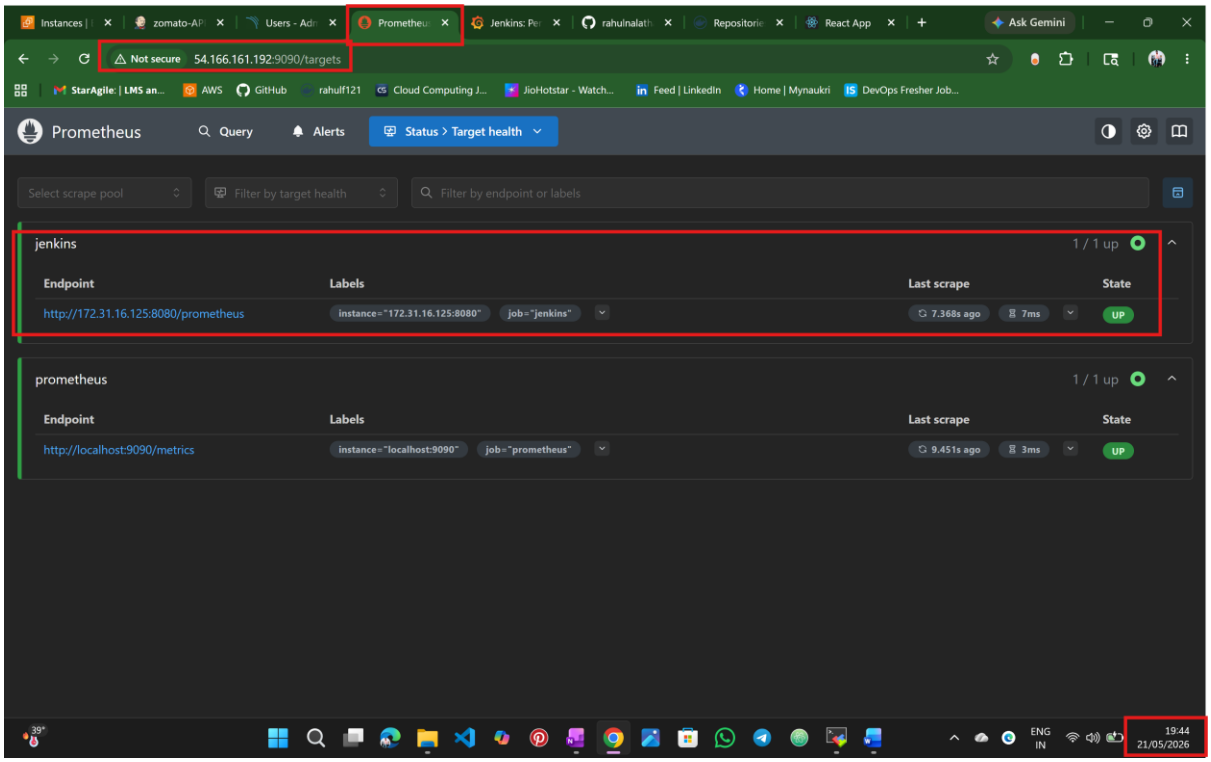
Step:12 we can see the through automation that docker image is pushed into docker hub



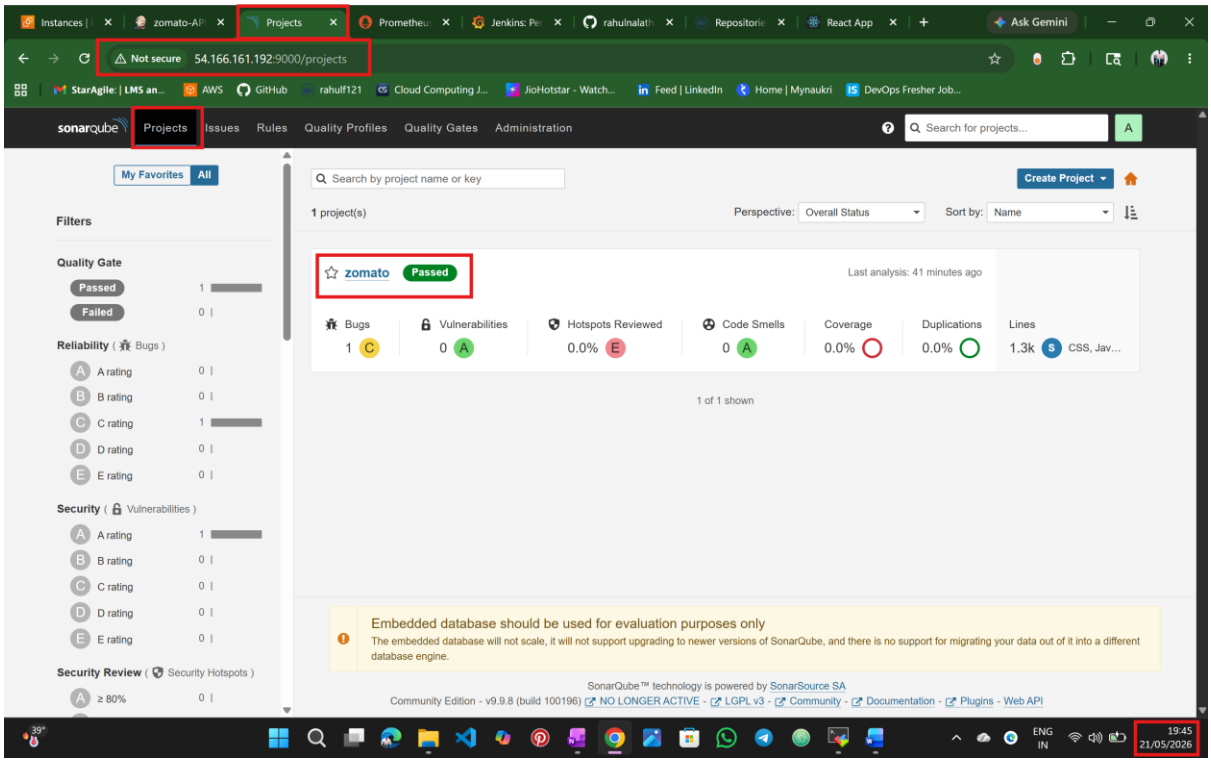
Step:13 finally we can access the application via k8s worker node ip address

Step:14 finally this is the visualization of Grafana.





Step:15 sonarQube code analysis



Nalathati Rahul